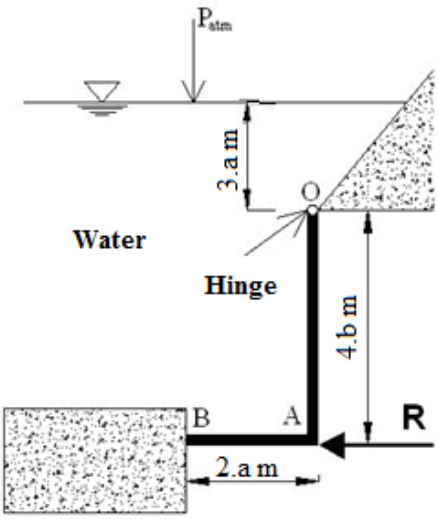
2020-2021 SPRING SEMESTER FLUID MECHANICS MIDTERM EXAM, Date:26/04/2021

Name: **Surname:** **Student ID:**

Question 3 (25p):

The rectangular OAB gate is hinged at point O and rests a wall at point B. What is the minimum force R required to hold the gate in place? The width of the gate is 3 m. Neglect the weight of the gate and the friction on the joint. The other side of the gate is in contact with the atmosphere.



a, and b are the last two digits of your student ID.
For example, Student ID: 17042082 a=8 b=2